What is `main.tex`?

Definition: Role of `main.tex`

`main.tex` is the **root file** of the project. It does two things:

- Sets up the **preamble**: loads packages, defines colors, macros, and environments.
- Pulls in **content files** with `\include{...}`.

You **never write slide content directly in** `main.tex` — only setup goes here.

The very first line sets the document type:

```
\documentclass[11pt, aspectratio=169, handout]{beamer}
```

- `beamer` — the presentation class
- `11pt` — base font size
- `aspectratio=169` — widescreen 16:9 slides
- `handout` — strips overlays/animations for printing
  - Omit `handout` for on-screen presentations with animations.
    - You can switch between modes by commenting/uncommenting this option.

Preamble: The Setup Zone

Everything between `\documentclass` and `\begin{document}` is the **preamble**.

The preamble in `main.tex` is organized into sections:

1. Encoding & typography
2. Beamer theme & navigation
3. Commonly used packages
4. Hyperlink settings
5. Custom colors
6. Frame color shortcuts
7. Colored box environments

Each section is marked with a `% --` comment header.

Loading Packages

Packages extend LaTeX's capabilities. Syntax:

```
\usepackage[options]{packagename}
```

Key packages in `main.tex`:

- `amsmath` — mathematical typesetting
- `xcolor` — custom colors
- `tcolorbox` — colored content boxes
- `fontawesome5` — icons (ℹ, ⚠)
- `tikz` / `pgfplots` — diagrams and graphs

Custom Colors

`main.tex` defines five named colors used throughout:

```
\colorlet{myblue}{black!35!blue}
\colorlet{myred}{black!35!red}
\colorlet{mygreen}{black!35!green}
\colorlet{mycyan}{cyan}
\colorlet{mygray}{black!65}
```

`black!35!blue` means: 35% black mixed into blue — giving a **darker, richer blue**.

Frame Color Shortcuts

To set the frametitle bar color, call one of these **before** `\begin{frame}`:

```
\blueheader   \redheader   \greenheader   \cyanheader   \grayheader
```

Example usage:

```
\blueheader
\begin{frame}{My Slide Title}
...content...
\end{frame}
```

Colored Content Boxes

Five theorem-style box environments are defined:

- `bluebox` — Definition
- `redbox` — Theorem
- `greenbox` — Example
- `cyanbox` — Exercise
- `graybox` — Explore

Usage requires a **title** and a **label**:

```
\begin{bluebox}{My Title}{mylabel}
...content...
\end{bluebox}
```

Focus Boxes

For **highlighted text with no title**, use focus boxes:

```
\begin{bluefocus}   \begin{redfocus}
\begin{greenfocus}   \begin{cyanfocus}
\begin{grayfocus}
```

Example — this sentence is in a blue focus box:

Focus boxes have a light colored background and no border or title.

Warning and Info Macros

Two icon macros for callouts:

- **Not part of the syllabus**
- **For your information**

Custom message versions:

```
\warnCustomMsg{Your message here}
\infoCustomMsg{Your message here}
```

Custom messages let you replace the default text with anything you like.

Including Content Files

Slide content lives in **separate** `.tex` files, included at the bottom of `main.tex`:

```
\include{Abstract-Linear-Algebra}
\include{French-Pronunciation}
\include{Italian-Pronunciation}
```

- Each file contains only frames — no preamble, no `\begin{document}`.
- Comment out a line to temporarily exclude that module from the build.
- `\include` always starts on a new page; use `\input` if you don't want that.

\newcommand — Defining Macros

```
\newcommand{\name}[n]{definition}
```

- `\name` — the command being created
- `[n]` — number of arguments (1–9); omit if none
- `definition` — the expansion; use `#1`, `#2` for arguments

From `main.tex` — no arguments:

```
\newcommand{\warn}{
\textcolor{orange}{\faExclamationTriangle \textbf{Not part of the
syllabus}}}
```

With 2 arguments, first has default `orange`:

```
\newcommand{\warnCustomMsg}[2][orange]{\textcolor{#1}{\textbf{#2}}}
```

`\tcbset` — Global Box Defaults

`\tcbset{key=value, ...}` sets **default options** inherited by all boxes that follow.

From `main.tex`:

`colback=white`	background color
`fonttitle=\bfseries`	bold title text
`breakable`	allow page breaks
`enhanced`	enable advanced drawing
`top=2pt, bottom=0pt`	inner vertical padding
`left=0pt, right=6pt`	inner horizontal padding
`boxsep=4pt`	space between border and content

Every `\newtcbtheorem` and `\newtcolorbox` inherits these automatically.

`\newtcbtheorem` — Numbered Box Environments

```
\newtcbtheorem[counter opts]{envname}{Title}{tcb opts}{prefix}
```

- `[counter opts]` — `use counter from=bluebox` shares one counter across all box types
- `envname` — name used in `\begin{...}`
- `Title` — text shown in box header, e.g. *Definition*
- `tcb opts` — extra style, e.g. `colframe=myblue`
- `prefix` — label namespace for `\ref`, e.g. `bluebox`

Usage:

```
\begin{bluebox}{My Title}{mylabel}
...content...
\end{bluebox}
```

Template Showcase: Overview

This presentation demonstrates the building blocks available in this template.

- **Header colors**: `\blueheader`, `\redheader`, `\greenheader`, `\cyanheader`, `\grayheader`
- **Theorem boxes**: `bluebox`, `redbox`, `greenbox`, `cyanbox`, `graybox`
- **Focus boxes**: `bluefocus`, `redfocus`, `greenfocus`, `cyanfocus`, `grayfocus`
- **Margin macros**: `\warn`, `\info`, `\warnCustomMsg`, `\infoCustomMsg`
- **Math macros**: `\norm{}`, `\abs{}`

Template Showcase: Blue

Definition: Definition

A **set** S is called *finite* if there exists a bijection $S \to \{1, \ldots, n\}$ for some $n \in \mathbb{N}$.

Definition 1: Definition (with label)

Same box, but numbered and referenceable via `def:finite`.

bluefocus — a titled-free highlight box. Use it to draw attention to a key formula or remark without a formal theorem label.

$$e^{i\pi} + 1 = 0$$

Template Showcase: Red

Theorem

For all $a, b \in \mathbb{R}$,

$$(a + b)^2 = a^2 + 2ab + b^2.$$

2: Theorem (numbered)

Same box with a counter shared across all box types.

redfocus — use for warnings, important notes, or anything that demands immediate attention.

Avoid using *red* purely decoratively; reserve it for critical content.

Template Showcase: Green

Example

Let $f(x) = x^2$. Then $f'(x) = 2x$, so the slope at $x = 3$ is

$$f'(3) = 6.$$

3: Example (numbered)

Same box, numbered.

greenfocus — ideal for worked examples, intuition, or positive reinforcement of a concept just introduced.

Template Showcase: Cyan

Exercise

Compute $\int_0^1 x^2\,dx$ and verify that the answer equals $\frac{1}{3}$.

4: Exercise (numbered)

Same box, numbered.

cyanfocus — well-suited for exercises, tasks, or student prompts that do not need a formal label.

Exploration

Pick any two vectors $u, v \in \mathbb{R}^2$. Compute $\|u + v\|$ and compare it with $\|u\| + \|v\|$. What do you notice?

5: Exploration (numbered)

Same box, numbered.

grayfocus — suited for open-ended exploration, background remarks, or supplementary context.

Template Showcase: Macros

Warning and info macros:

- ⚠ **Not part of the syllabus**
- ℹ **For your information**
- ⚠ **Custom warning in any color**
- ℹ **Custom info in any color**

Math macros:

- Norm: $\|v\| = \sqrt{\langle v, v\rangle}$
- Absolute value: $|x - y| \leq |x| + |y|$
- Combined: $|\langle u, v\rangle| \leq \|u\| \, \|v\| \quad$ (Cauchy–Schwarz)

Abstract Linear Algebra: Vector Space

A **vector space** over a field $\mathbb{F}$ is a set V equipped with

- vector addition: $\boldsymbol{+ : V \times V \to V}$,
- scalar multiplication: $\boldsymbol{\cdot : \mathbb{F} \times V \to V}$,

satisfying for all $u, v, w \in V$, $\alpha, \beta \in \mathbb{F}$:

1. $u + v = v + u$ (commutativity)
2. $(u + v) + w = u + (v + w)$ (associativity)
3. There exists $0 \in V$ such that $v + 0 = v$ (additive identity)
4. For each v, there exists $-v$ with $v + (-v) = 0$ (additive inverse)
5. $\alpha(u + v) = \alpha u + \alpha v$ (distributivity I)
6. $(\alpha + \beta)v = \alpha v + \beta v$ (distributivity II)
7. $(\alpha\beta)v = \alpha(\beta v)$ (associativity of scalar mult.)
8. $1 \cdot v = v$ (unit property)

Remark

A vector space by itself only has the structure provided by addition and scalar multiplication.

- It does *not* include a notion of **length**, **angle**, or **orthogonality**.
- These geometric notions are **induced** once we add an **inner product** and, from it, a norm.

Abstract Linear Algebra: Inner Product

Definition: Definition

An **inner product** on a real vector space V is a function

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$$

such that for all $u, v, w \in V$, $\alpha \in \mathbb{R}$:

1. $\langle u, v \rangle = \langle v, u \rangle$ (symmetry)
2. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ (linearity in the first slot)
3. $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$ (homogeneity)
4. $\langle v, v \rangle \geq 0$, and $\langle v, v \rangle = 0 \iff v = 0$ (positivity/definiteness)

Definition

Given an inner product $\langle \cdot, \cdot \rangle$ on V, the induced **norm** is

$$\|v\| = \sqrt{\langle v, v \rangle}, \quad v \in V.$$

This norm yields geometric notions:

- **Length**: $\|v\|$
- **Angle**: $\cos\theta = \dfrac{\langle u, v \rangle}{\|u\| \, \|v\|}$ (for $u, v \neq 0$)
- **Orthogonality**: $u \perp v \iff \langle u, v \rangle = 0$

Abstract Linear Algebra: Example

Example

Let $V = \mathbb{R}^2$ with the standard inner product

$$\langle u, v \rangle = u_1 v_1 + u_2 v_2 .$$

For $u = (3, 4)$ and $v = (4, -3)$:

$$\langle u, v \rangle = 3 \cdot 4 + 4 \cdot (-3) = 12 - 12 = 0.$$

Thus $u \perp v$. Also $\|u\| = \sqrt{3^2 + 4^2} = 5$.

Exercise

Let $u = (1, 2, 2)$ and $v = (2, 0, 1)$ in $\mathbb{R}^3$ with the standard inner product.

1. Compute $\langle u, v \rangle$.
2. Find $\|u\|$ and $\|v\|$.
3. Determine the cosine of the angle between u and v.

Abstract Linear Algebra: Exploration

Task for Exploration

Work with the standard inner product on $\mathbb{R}^2$ or $\mathbb{R}^3$.

1. Pick two vectors u and v. Compute
$$|\langle u, v\rangle|, \quad \|u\|, \quad \|v\|, \quad \|u+v\|.$$
2. Compare $|\langle u, v\rangle|$ with $\|u\| \, \|v\|$. Try several pairs.
3. Compare $\|u+v\|$ with $\|u\| + \|v\|$. Does it always hold?
4. Explore special cases: u and v linearly dependent; $u \perp v$.

You are rediscovering the Cauchy–Schwarz inequality and the triangle inequality. Equality holds when u and v are linearly dependent (Cauchy–Schwarz), and when u and v point in the same direction (triangle).

Abstract Linear Algebra: Beyond the Curriculum

⚠ Not part of the syllabus

Linear algebra has powerful real-world applications:

- **Cryptography**: RSA and lattice methods use vector spaces and linear maps.
- **Engineering**: Signal processing and compression use inner products and orthogonal bases.
- **Machine Learning**: Vector spaces model data; norms and inner products measure similarity.